Detection of peptide using the Biuret Test

Purpose

To observe the color of the reaction between tryptone, glycine, and water with alkaline copper. To understand how the chemical reaction of peptide structure affects the copper.

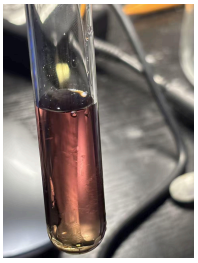
Material

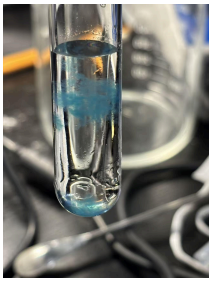
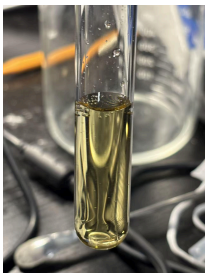
Tryptone, glycine, copper(II) sulfate solution, sodium hydroxide solution, distilled water, test tubes, a test-tube rack, pipettes

Procedure

- Add tryptone to A and D, Glycine to B, and distilled water to C
- Add sodium hydroxide solution to each tube.
- Add copper sulfate to A, B, and C. Add the same amount of water to D.

Observation

Tube	Composition	
A	Tryptone, NaOH, Copper Sulfate	

B	Glycine, NaOH, copper	
C	Water, NaOH, copper	
D	Tryptone, NaOH	

Chemical explanation

In an alkaline environment, the N and H of the peptide bond lose protons, causing the original nitrogen atom to pair with Cu2. Cu can form a coordinate bond with four nitrogen atoms, forming a chelate complex - there is a central ion that accepts the surrounding ligands' electrons to form a coordination bond (sharing two electrons). In this chelate complex, there will be two or more atoms and ligands connected. The deprotonated nitrogen atom becomes a strong ligand that affects the absorption of light. Tryptone itself has a yellow color. The color of A comes from the copper and peptide complex, B because it lacks a peptide bond, causes the combination of oxygen and copper forms copper glycinate, which is clear blue. C because there are no ligands, forms a precipitate. Therefore, the overall reaction comes from the coordination bonds between the surrounding ions and the central particle.